elabortate expand on Semester faculty and teacher OS capppablties and also Semester intlgense providing them there own exectutive assistent reasearch partner and TA

Semester's Faculty and Teacher OS should give each educator one connected workspace for teaching, research, administration, student support, and professional work—with Semester Intelligence acting as their **chief-of-staff** layer: an executive assistant, research partner, and bounded AI teaching assistant.

These should be three coordinated modes of the same intelligence system, not three disconnected chatbots. They share the educator's authorized context, but operate with different permissions, source collections, and decision boundaries.

Your existing Course Studio design already includes week planning, student preview, assignments, rubrics, accessibility checks, course AI rules, teaching-team context, and controlled grade release. The expansion below builds on that foundation. Those artifacts are design/prototype evidence—not proof of deployed teaching, research, or agent functionality. [1] [2]

1. What the Faculty OS should encompass

Faculty work extends beyond delivering course content. Semester should connect six work domains while keeping their records and permissions separate.

Domain	Faculty capabilities	Intelligence support
Teaching	Course design, instruction, assignments, assessments, feedback, office hours	Prepare materials, explain patterns, draft feedback, answer approved course questions
Research	Literature, projects, methods, analysis, writing, collaboration, reproducibility	Search and compare sources, critique methods, assist with code, maintain evidence
Administration	Meetings, deadlines, service commitments, forms, approvals	Briefings, prioritization, preparation, follow-through
Student support	Appointments, referrals, accommodations implementation, mentoring	Prepare permitted context, draft outreach, route unresolved questions
Teaching-team management	TA assignments, grading calibration, delegation, review	Queue organization, workload summaries, consistency checks
Professional development	Teaching portfolio, publications, grants, annual-review evidence	Organize artifacts, draft summaries, track verified contributions

The existing platform map already includes faculty course lifecycle, research and writing workspaces, assessment, teaching-team delegation, academic records, and shared workflow engines. Research operations and faculty professional-development functions below are proposed deeper specifications, not verified completed modules. [3] [4]

Support different educator contexts

Use one Faculty OS with configurable capabilities for:

- University faculty and lecturers.
- Adjunct and visiting instructors.
- Graduate instructors.
- Teaching assistants.
- Lab and studio instructors.
- Clinical or placement educators.
- Program directors and department chairs.

If Semester expands into K–12 teacher workflows, treat that as a separately reviewed institutional configuration: family access, minors' data, classroom operations, and district authority should not simply inherit university defaults.

2. Faculty home: a daily command center

The home screen should answer:

> What needs my attention, what is ready for review, and what can Semester prepare for me?

Proposed screen structure

```
Faculty Home
├── Today
│   ├── Teaching and meetings
│   ├── Deadlines and conflicts
│   └── Briefing
├── Action Center
│   ├── My work
│   ├── Review and approve
│   ├── Waiting on others
│   └── Exceptions
├── Courses / Course Studio
├── Assessment and feedback
├── Teaching team
├── Student appointments and referrals
├── Research workspace
├── Writing and publications
├── Grants and project administration
├── Service and committees
```

```
├── Professional portfolio
└── Intelligence and permission settings
```

These are proposed navigation groups, not asserted existing routes.

What the daily briefing contains

A useful briefing could show:

- Teaching sessions and preparation needed.
- Unreleased feedback and grades.
- Questions awaiting a human answer.
- Assignment drafts blocked by accessibility or policy issues.
- Research deadlines and collaborator dependencies.
- Meetings requiring preparation.
- Integration failures affecting rosters, calendars, or grade passback.
- Proposed next actions with their sources.

Every item should distinguish:

```
Observed fact → generated interpretation → suggested action → approved action → confirmed outcome
```

The existing Course Studio already models loading, empty, and source-error states, including preserving drafts when the external course source cannot load. Extend that resilience to the faculty-wide home. [1]

3. Course Studio: complete teaching operations

Course design and preparation

Faculty should be able to:

- Import or author a syllabus.
- Define learning objectives.
- Map readings, activities, and assessments to objectives.
- Build a week or unit plan.
- Compare planned workload across assignments.
- Set office hours and communication expectations.
- Define grading schemes and rubric versions.
- Configure course and assignment AI rules.
- Review accessibility before publication.
- Preview the actual student experience.

- Reuse a prior course without carrying forward students or obsolete permissions.

The current master map names syllabus authoring, objectives, resources, assignment and assessment builders, question banks, rubrics, accessibility review, AI rules, and course archive/export.[4]

Publication workflow

```
Faculty intent
  → approved source selection
  → draft content
  → objective and assessment alignment
  → accessibility review
  → AI/integrity rules
  → student preview
  → faculty approval
  → publication
  → calendar / Action Center / notifications update
  → confirmation and history
```

A published item should have a version, effective date, audience, responsible author, and change history.

A later deadline change should preview:

- Which students and sections are affected.
- Whether individual approved due dates need recomputation.
- Which calendars and reminders will change.
- Whether already-submitted work requires special handling.
- What communication will be sent.

Assessment and feedback

Complete the workflow beyond a gradebook table:

1. Author instructions and rubric.
2. Configure attempts, deadlines, accommodations, and allowed AI assistance.
3. Collect submissions with receipts.
4. Assign grading work to authorized humans.
5. Prepare feedback and proposed rubric annotations.
6. Review and calibrate grading.
7. Approve release.
8. Pass back to the authoritative system where applicable.
9. Reconcile accepted, rejected, and pending records.
10. Process regrades and corrections separately.

Your Course Studio prototype already distinguishes draft content from student-visible material and identifies an external gradebook as authoritative until cutover. It also includes an accessibility publishing gate and controlled grade-release UI. The prototype's simulated release receipt is not evidence of a real backend transaction. [1]

4. Semester Intelligence as an executive assistant

The executive assistant manages coordination and preparation—not the educator's academic authority.

Core capabilities

Capability	Prepared output	Action boundary
Daily briefing	Prioritized work with evidence and dependencies	Faculty chooses priorities
Calendar planning	Proposed teaching, research, and meeting blocks	Calendar writes require approved scope
Meeting preparation	Agenda, source-linked briefing, open decisions	Recording/transcription requires appropriate permission
Follow-through	Draft tasks and reminders from confirmed decisions	No inferred commitment becomes official automatically
Communication drafting	Student announcements, collaborator updates, administrative replies	Recipient and content review before sending
Deadline coordination	Grant, grading, committee, reporting checklist	Uses authoritative dates; flags uncertainty
Workload planning	Conflicts, capacity scenarios, proposed deferrals	Transparent assumptions; no punitive evaluation
Administrative preparation	Forms and evidence packages	Authorized human signs/submits consequential records

Example: preparing a faculty week

Faculty request:

> "Prepare my week. Preserve two research blocks, prepare Thursday's lecture, and organize the grading backlog."

Semester should:

1. Read only calendars, courses, projects, and queues within the granted scope.
2. Identify fixed commitments and recorded deadlines.
3. Ask for missing constraints only when needed.
4. Produce a proposed schedule.
5. Link each suggested task to its originating course, project, or meeting.
6. Show tradeoffs and unresolved dependencies.
7. Preview any calendar changes.

8. Apply approved changes and return receipts.

It should not silently cancel office hours, move assessment dates, email students, or disclose private research plans.

Faculty-controlled memory

Separate:

- Personal preferences: preferred working hours, writing style, briefing format.
- Course context: published rules and permitted materials.
- Research context: project-specific sources and decisions.
- Institutional records: official data with institutional retention and access rules.

Faculty should be able to inspect, correct, delete where permitted, and disable personal memory. A course relationship ending must not leave student context available through persistent assistant memory.

5. Semester Intelligence as a research partner

The research partner should help faculty work from evidence, not generate plausible-looking scholarship.

Research workspace

Proposed screens:

- Project overview and research questions.
- Source library.
- Literature evidence matrix.
- Methods and analysis plan.
- Data-access and governance record.
- Code and computational notebook workspace.
- Results and reproducibility.
- Manuscript and revision workspace.
- Collaborator tasks and decisions.
- Grant and dissemination tracker.

Literature workflow

```
Research question
  → search scope and inclusion criteria
  → candidate sources
  → duplicate and relevance review
  → evidence extraction
  → claim-to-source matrix
```

```
→ synthesis
→ researcher review
→ manuscript or project artifact
```

Each extracted finding should identify:

- Source and exact supporting location where available.
- Study type and population.
- Methods.
- Reported result.
- Limitations.
- Whether Semester read the full text or only metadata/abstract.
- Whether a statement is reported by the author or inferred by the assistant.

Research capabilities and limits

Task	Useful assistance	Required limit
Literature exploration	Query design, source comparison, contradictory findings	No invented citations; no full-text claim without access
Methods design	Alternatives, assumptions, confounders, feasibility questions	Researcher owns methodological decisions
Code and analysis	Draft code, debugging, test design, reproducible execution	Label generated versus executed results
Data preparation	Validation proposals, schema review, transformation drafts	Preserve raw data; review consequential transformations
Manuscript development	Outline, clarity, evidence-gap review, revision tracking	Preserve authorship accountability and disclosure requirements
Grant preparation	Source-grounded requirements and draft narratives	No fabricated preliminary results, personnel, budgets, or approvals
Peer-review response	Comment-to-change matrix and evidence references	Reviewer material remains within permitted confidentiality boundaries
Reproducibility	Versions, parameters, dependencies, provenance, rerun records	Successful reproducibility requires actual execution evidence

Example: source-supported synthesis

Faculty request:

> “Compare these twelve papers and identify disagreements about the mechanism.”

The output should be a matrix—not merely an essay:

Claim	Supporting papers	Contradicting papers	Methods differences	Uncertainty	Next investigation

If evidence is insufficient, Semester should say so. “Research gap” is a hypothesis requiring review, not a fact established by a fluent summary.

Sensitive research boundary

The inspected AI specification currently calls for no student records to AI providers and identifies the intelligence layer as design-stage. Do not quietly expand that policy to accommodate research or individualized grading. Restricted participant data, unpublished confidential material, or student-linked datasets require a separately authorized processing design and evidence-backed controls. [2]

Algorithm governance also excludes sensitive health data, private messages, accommodation details, disciplinary records, and other sensitive inputs without explicit policy or consent. Consent alone should not be treated as permission to bypass institutional restrictions. [5]

6. Semester Intelligence as an AI TA

The AI TA should support instruction without impersonating a human TA or becoming the instructor of record.

Student-facing capabilities

Within approved course and assignment rules, it can:

- Explain concepts from instructor-approved materials.
- Answer syllabus and logistics questions.
- Offer hints and staged practice.
- Ask diagnostic questions.
- Explain feedback already released.
- Generate non-assessed practice activities.
- Direct students to office hours or support.
- Escalate unresolved or ambiguous questions.

Your existing prototype already distinguishes assignment modes such as “Off,” “Explain only,” and “Feedback on drafts.” It also specifies that course rules cannot be more permissive than institutional rules. Extend this into an enforceable policy model rather than a display-only setting. [1]

Faculty-facing capabilities

- Group recurring questions by concept.
- Draft FAQ entries.
- Suggest a clarification announcement.
- Prepare discussion prompts.
- Draft practice problems and answer keys in instructor-only space.

- Suggest rubric-aligned feedback.
- Highlight inconsistent feedback for human review.
- Organize grading and regrade queues.
- Prepare office-hours materials.
- Summarize permitted aggregate learning signals.

Keep assessment materials separated

Use distinct source collections:

1. Student-visible teaching materials.
2. Instructor-only answer keys and assessments.
3. Authorized teaching-team materials.
4. Student submissions and official records.
5. Student-private notes and AI conversations.

A student-facing retrieval request must never access an answer key because the same faculty assistant can access it.

Faculty do not automatically gain access to students' private study conversations. Faculty-facing summaries should use approved, purpose-limited reporting—not covert surveillance.

Example: recurring student question

1. Students ask about a concept using approved materials.
2. Semester answers within the assignment policy.
3. Unresolved questions enter a faculty review queue.
4. The faculty view shows a permitted aggregate theme, not private transcripts by default.
5. Semester drafts a clarification with source links.
6. Faculty approves publication.
7. Student-visible materials update with version history.

Grading assistance is not grade authority

AI may propose feedback or annotations under an approved processing policy. It should not independently assign official grades, release them, deny a regrade, or infer misconduct.

Any future student-submission analysis must first resolve the current no-student-records-to-AI constraint. Until then, teaching assistance can operate on approved course content, synthetic examples, and appropriately authorized non-student-specific materials.[2]

7. Human TA and AI TA teamwork

Semester should support human teaching teams more effectively, not obscure their accountability.

Human role	Proposed authority	AI support
Instructor of record	Course policy, assessment approval, final release	Preparation, evidence, drafts, consistency checks
Co-instructor	Institution-approved shared teaching responsibilities	Coordination within assigned scope
Graduate TA	Assigned discussions, office hours, draft grading	Queue preparation and source-grounded assistance
Undergraduate assistant	Narrower, explicitly granted functions	Approved logistics and practice support
Lab/studio staff	Assigned sessions and relevant operational tasks	Preparation checklists and instructional resources

Teaching-team workflow

```
Provision relationship
  → grant scoped capabilities
  → complete required training
  → assign work
  → calibrate
  → draft feedback / grading
  → review exceptions
  → authorized release
  → reconcile
  → sunset term-specific access
```

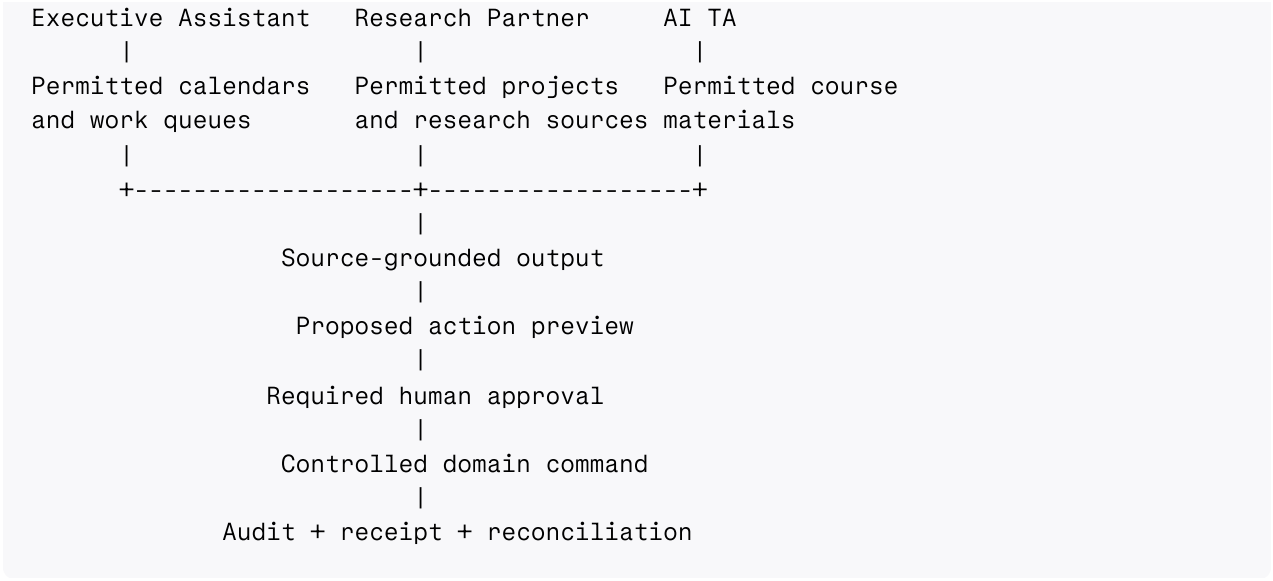
A TA should see only relevant sections, submissions, and permitted accommodation instructions. Their scope should not automatically include private advising cases, research projects, institutional finance, or official record corrections.

The existing role specification already calls for course-scoped TA permissions and access sunset when the teaching assignment ends. [3]

8. One gateway, three modes, explicit permissions

A proposed architecture:

```
Faculty request
      |
Context and intent selection
      |
Tenant + role + relationship + policy + classification checks
      |
      +-------------------+------------------+
      |                   |                  |
```



This matches your intelligence specification's core principle: person, role, tenant, course, policy, source authority, and consent determine allowed models and tools. Evaluation, correction, audit, cost controls, and human escalation belong in the same gateway. [2]

Required capability record

Each assistant skill should define:

```
Purpose
Permitted roles
Resource relationships
Allowed sources
Excluded data
Allowed tools
Approval requirements
Output and provenance
Evaluation cases
Failure and escalation behavior
Budget / runtime limits
Memory rules
Audit and retention
Activation gate
Operating owner
```

For example, `draft_announcement` and `publish_announcement` should be separate capabilities. Being allowed to prepare content does not grant permission to send it.

Avoid a misleading confidence meter

Instead of presenting a single opaque "92% confident" score, show:

- Which sources were used.
- Whether they are official or provisional.
- Whether the answer is directly supported or inferred.

- What information is unavailable.
- Which claims need human confirmation.

Your algorithm-governance specification requires explanations, uncertainty, correction paths, versioning, and human review for consequential outputs. [5]

9. Faculty screens to specify next

Screen	Essential functions	Completion evidence
Faculty Today	Briefing, priorities, commitments, blockers	Source freshness and permission tests
Review and Approve	Exact draft, recipient/target preview, impact, expiry	Approval binding and replay tests
Course Studio	Syllabus, weekly plan, student preview, publication	Draft/published separation and update propagation
Course AI Rules	Policy modes, sources, tools, assignment exceptions	Enforcement at model/tool invocation
Teaching Team	Relationships, delegation, assignment, expiry	Scope and revocation tests
Assessment Review	Rubrics, feedback, moderation, release	Human authority and reconciliation
Course Questions	Approved FAQ, unanswered queue, escalation	Answer-key isolation and integrity tests
Office Hours	Booking, preparation, follow-up	Calendar conflicts and visibility tests
Research Project	Evidence, methods, tasks, versions	Source and collaborator isolation
Literature Matrix	Claim/source comparison, extraction, limitations	Citation and grounding evaluation
Analysis Workspace	Code, runs, inputs, outputs, reproducibility	Execution receipts and sandbox controls
Professional Portfolio	Teaching/research/service evidence, exports	Verified provenance and sharing controls
Intelligence Settings	Permissions, sources, memory, budgets, audit	Revocation, retention, and kill-switch tests

These are proposed specifications. Reconcile them with existing screens before adding new routes.

10. Implementation priorities and activation gates

First: complete existing teaching workflows

The gap register already identifies faculty AI-policy publication, roster management, and class communication as unmapped steps. Those are concrete reconciliation targets before expanding into a large new research-agent surface. [6]

Then: prove three complete assistant workflows

1. Executive assistant: prepare a daily briefing and one approved calendar change.
2. Research partner: produce a verified source-comparison matrix with explicit access limitations.
3. AI TA: answer a course question, refuse a prohibited assessed-answer request, and escalate ambiguity.

Required owners and gates

Workstream	Proposed owner	Activation evidence
Faculty workflow	Faculty product lead + faculty reviewers	End-to-end teaching acceptance
Research partner	Research product lead + institutional research/privacy owners	Source grounding, confidentiality, execution controls
AI TA	Learning lead + instructors + AI governance	Integrity-policy and answer-key isolation tests
Gateway and tools	Platform/AI engineering	Authorization, revocation, budget, audit, failure tests
Grade authority	Registrar/academic owner	Institution-specific approval and reconciliation
Accessibility	Design/accessibility lead	Tested critical journeys
Operations	Engineering operations + implementation	Monitoring, incident, rollback, support readiness

The current intelligence document says gateway logic and tests are designed/drafted, the red-team suite has not been run, and staging red-team execution is the next action. Faculty-agent expansion should not bypass that gate. [2]

The strongest Faculty OS is therefore not “AI that does a professor’s job.” It is a connected workspace that removes administrative friction, strengthens evidence-based teaching and research, helps human teams operate coherently, and leaves academic judgment, sensitive decisions, and official authority with the appropriate people.

⁂

1. The Main Semester design system (2) copy/handoff/design_handoff_permissions_and_queues/design/templates/CourseStudio.dc.html
2. The Main Semester design system (2) copy/docs/platform/AI_INTELLIGENCE_PLATFORM.md
3. The Main Semester design system (2) copy/handoff/specifications/depect-semester-flow-for-every-role.pdf
4. The Main Semester design system (2) copy/uploads/Map semester out as a whol provide full map out of.pdf
5. The Main Semester design system (2) copy/docs/system/algorithms/SEMESTER_ALGORITHM_GOVERNANCE.md
6. The Main Semester design system (2) copy/docs/master/SEMESTER_GAP_REGISTER.md
7. https://www.perplexity.ai/search/587c98eb-35ce-4508-ac86-259785dc1b94

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   copy/docs/system/algorithms/SEMESTER_ALGORITHM_CATALOG.md
9. https://www.perplexity.ai/search/e6152e25-b4ea-4159-bb88-db3cc83f184b
10. https://www.perplexity.ai/search/f1fd8531-df7e-42c1-8e27-df383822c5b1